

Colton Carroll

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Education

University of Washington | Bachelors in Computer Engineering | GPA: 3.9 Expected June 2027

Relevant Coursework: Data Structures and Parallelism, Systems Programming, Operating Systems, Distributed Systems, Hardware-Software Interface, Linux and the CLI, Statistics, Linear Algebra

Experience

Software Development Engineer Intern June 2025 – September 2025
Amazon Alexa | Seattle, WA

- Owned a full-stack project for the creation & management of customer-created home automation groups. These allow users to organize their smart home automations intuitively.
- Added 3 new interactive pages in the Alexa app using React Native for group management.
- Created 4 APIs in the backend with caching, Dynamo DB storage, and latency monitoring.
- Interfaced often with product, UX, and software teams to build a production-ready project.

Firmware Subteam Lead October 2024 – Current
Washington Superbike Club | University of Washington

- Lead a team to write, test, and maintain firmware for a custom-built electric motorcycle.
- Organize weekly meetings and work parties to keep the team on schedule.
- Implement new features, like hardware in the loop testing, to prepare for competition.

Undergraduate Researcher September 2024 – Current
Advancing Movement and Performance Lab | University of Washington

- Programmed a microcontroller to interface over Bluetooth with a wearable fitness tracker to analyze and act upon cardiovascular indicators in patients with spinal cord injuries.
 - Patients participating in the clinical trial saw an improvement in their quality of life as a result of my device's synthesis with their spinal electrical stimulation therapy regimen.
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Technical Projects

Custom Macropad Keyboard March 2024

- Wrote software for the keyboard based off of the CircuitPython and KMK python libraries.
- Designed a custom circuit board and 3d-printed enclosure using Altium and Autodesk.
- Tested and soldered microcontroller, LEDs, and SMD components onto the PCB.

Wi-Fi Controlled Computer Lighting January 2023

- Built custom PC lighting components with an ESP32 controller on a custom designed PCB.
 - Enabled Wi-Fi connectivity, automation, and custom lighting control from a mobile app.
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Skills

- Proficient in Java, Python, TypeScript, React, C, and C++
- Designing fault-tolerant systems, working in fast-moving teams, passion for building
- Experienced with Linux, Bash scripting, shell commands, and electronics design